PK/PD Models

A Case Study in Model Reproducibility and Transparency

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What Are PK/PD Models?

- **PK/PD models** link **drug dose** $\rightarrow$ **body response**. Address the effect of drug exposure and response.
- **Pharmacokinetics (PK):** what the body does to the drug — absorption, distribution, and clearance.
- **Pharmacodynamics (PD):** what the drug does to the body — physiological effect.
- Used to predict dose–response, guide clinical design, and improve efficiency [1].

$\boxed{\textbf{Dose}} \Rightarrow \boxed{\textbf{Plasma Concentration (PK)}} \Rightarrow \boxed{\textbf{Response (PD)}}$

Pharmacokinetics: Core Concepts

- **One-compartment:** assumes uniform drug distribution; concentration declines monoexponentially [2].
- **Multi-compartment:** accounts for fast (distribution) and slow (elimination) phases as tissues absorb differently [2].
- **In this model:** Berger & Rodbard (1989) used a one-compartment PK framework with a subcutaneous absorption step and first-order clearance.

Introduction to Methodology

Overview of Approach

1. **Model Reconstruction:** Rebuilt the classic PK/PD model from Berger & Rodbard (1989) [3].
2. **Numerical Implementation:** Solved coupled differential equations in Python using SciPy's ODE solver.
3. **Simulation:** Generated dynamic time-course outputs for multiple insulin formulations.
4. **Validation:** Compared simulated results against the original published data.
5. **Analysis:** Conducted stability and parameter-sensitivity tests to assess model behavior.
6. **Beyond the Model:** Interpreted results for reproducibility and real-world translation.

Each step builds toward reliable, reproducible modeling of insulin–glucose dynamics.

Case Study: Berger & Rodbard (1989)

- Based on ”*Computer Simulation of Plasma Insulin and Glucose Dynamics After Subcutaneous Insulin Injection*”.
- Simulates various insulin formulations and treatment variables — dose, timing, and meal patterns.
- Uses a one-compartment PK model with first-order elimination and a subcutaneous absorption step.
- **Goal:** rebuild the model from the paper alone to test reproducibility and transparency.

$\boxed{\textbf{Subcutaneous Depot}} \Rightarrow \boxed{\textbf{Plasma Insulin}} \Rightarrow \boxed{\textbf{Glucose Response}}$

Insulin and Glucose Dynamics

Key Hormones and Organs

- **Glucose:** Primary sugar used by cells for energy
- **Insulin:** regulates blood glucose by promoting uptake/downtake into the cell
- **Glucagon:** raises blood glucose by signaling the liver to release glycogen stores.
- **Pancreas:** produces both insulin and glucagon for glucose regulation.
- **Liver:** stores glucose as glycogen and releases it when signaled by glucagon.

Together, insulin and glucagon maintain blood-glucose homeostasis.

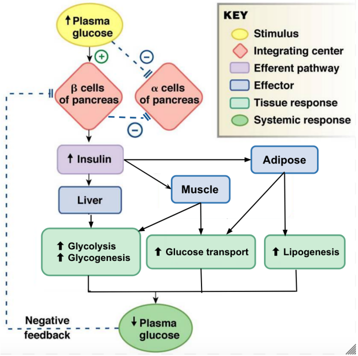


Figure: Overview of insulin response to hyperglycemia [4]

Time Course of Insulin Absorption

- Models how much insulin remains at the **injection site** over time. A(t) the percentage of injected insulin dosage remaining at the site of absorption after t seconds. The variable s is used to characterize the time course.
- The absorption curve is shaped by:
  - $A(t) = 100 - \dfrac{100t^s}{(T_{50})^s + t^s}$
  - $T_{50}(D) = aD + b$
- s: controls how steeply absorption occurs.
  T_{50}: time when 50% of the dose is absorbed (depends on dose D).
- Short-acting insulin $\rightarrow$ smaller T_{50}; long-acting $\rightarrow$ larger T_{50}.

$$\boxed{\textbf{Depot}} \Rightarrow \boxed{\textbf{Plasma}}$$

Fast absorption = early peak; slow absorption = extended effect.

Insulin Absorption

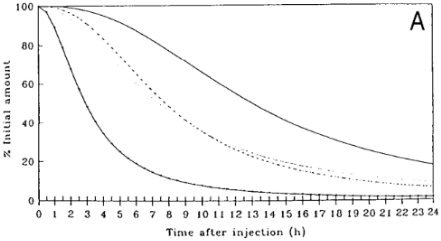


Figure: Graph A shows the predicted time course of insulin absorption using Equations 1 and 2 [3]

- Regular (Solid line with points)- short acting insulin
- NPH (dotted line) - intermediate acting
- Lente (dashed line) - intermediate acting
- Ultralente (solid line with no points)- long acting

Plasma Insulin Dynamics

- After absorption, insulin molecules move from the subcutaneous depot into the blood plasma. The change in plasma insulin concentration over time can then be understood as the balance between two opposing processes:
  - **Absorption rate:** insulin entering the bloodstream from the depot.
  - **Elimination rate:** insulin cleared from plasma at rate constant k_e.
- The net difference between these processes gives the rate of change of plasma insulin, showing whether its concentration is increasing or decreasing at that point in time:

$$\frac{dA}{dt} = \text{Absorption} - k_e A$$

- Plasma concentration is then:

$$I(t) = \frac{A(t)}{V_1}$$

  where V_1 is the distribution volume.
- This produces the plasma insulin curves that follow each absorption profile.

Plasma Insulin Profiles

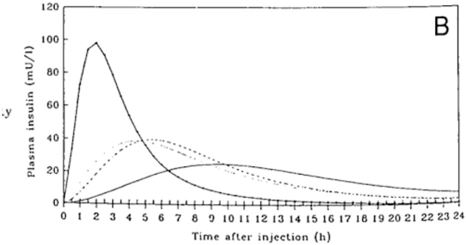


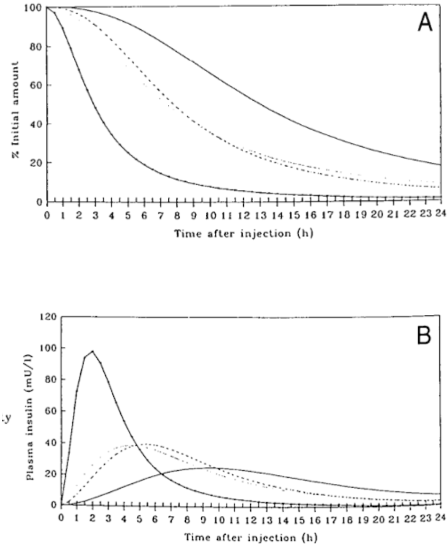
Figure: Simulated plasma insulin curves for four insulin formulations.

Simulation Parameters

- **Regular** — Solid line with points
  $s = 2.0,\ a = 0.05\ (h/U^{-1}),\ b = 1.7\ h$
- **NPH** — Dotted line
  $s = 2.0,\ a = 0.18\ (h/U^{-1}),\ b = 4.9\ h$
- **Lente** — Dashed line
  $s = 2.4,\ a = 0.15\ (h/U^{-1}),\ b = 6.2\ h$
- **Ultralente** — Solid line (no points)
  $s = 2.5,\ a = 0.00,\ b = 13\ h$

Short-acting insulins peak early;
long-acting are flatter and longer-lasting.

Figures


A
% Initial amount
100
80
60
40
20
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Time after injection (h)
B
Plasma insulin (mU/l)
120
100
80
60
40
20
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Time after injection (h)

Model Recreation

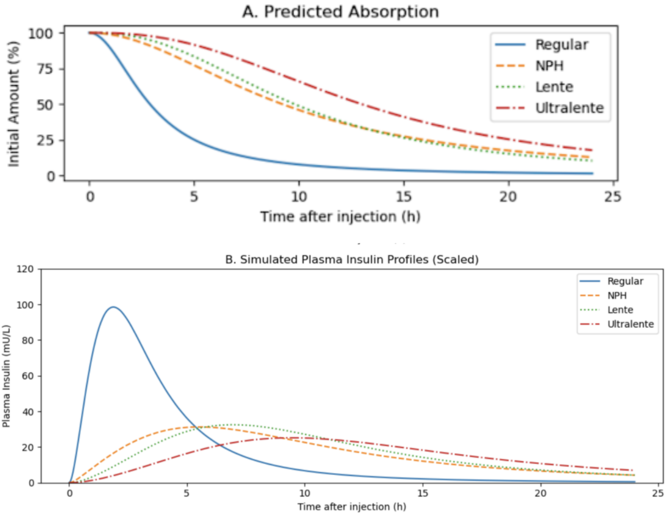
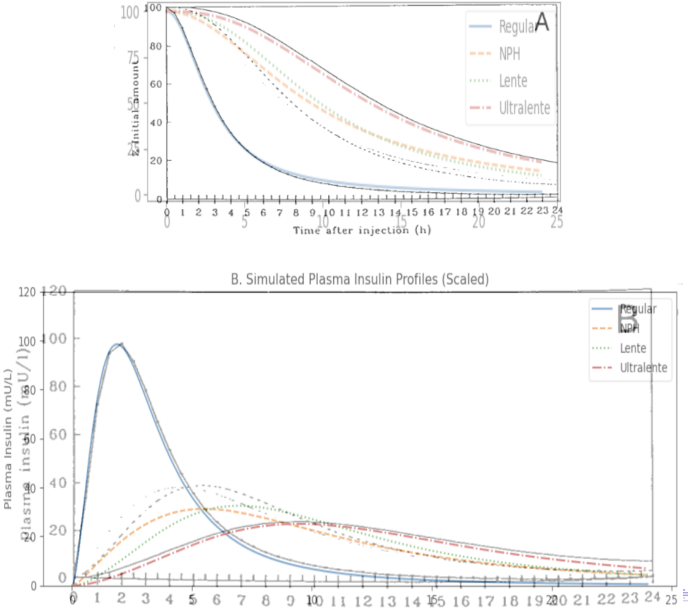


Figure: Simulated Graph

Comparison



A
Regular
NPH
Lente
Ultralente
% initial amount
Time after injection (h)
B. Simulated Plasma Insulin Profiles (Scaled)
B
Plasma Insulin (mU/L)
Plasma insulin (mU/l)

Why My Simulation Differs Slightly

Expected Numerical Differences

- **Parameter precision:** The 1989 paper rounded constants (k_e, a, b, s), and small changes in these values shift curve height or timing.
- **Solver method:** Modern adaptive ODE solvers (Runge–Kutta, LSODA) differ from the fixed-step schemes used in the original study.
- **Scaling / normalization:** The paper plotted normalized concentrations; I applied a constant scaling factor to match their published scale.

- Minor deviations are typical in model replication and highlight the importance of transparency and parameter reporting for reproducibility.

Same dynamics — slightly different numbers. What's important is that the model's shape, timing, and dynamics are consistent with the original.

Stability Analysis

Concept

Assesses how the insulin–glucose system responds to small disturbances using **linearization** [5].

- Determines whether glucose levels return to equilibrium (**homeostasis**) or diverge (**instability**, e.g., hypo-/hyperglycemia).
- In this model, the system tends to return to steady state following perturbations such as meals or insulin injections.

Stable System:	Returns to baseline after disturbance.
Unstable System:	Deviates and amplifies over time.

Parameter Sensitivity Analysis

Concept

Identifies which parameters most influence model outputs by varying one parameter at a time while keeping others constant [6].

Key Parameters

- s — Absorption rate shape: controls how steeply insulin is absorbed.
- a — Dose-response slope: how much dose affects absorption speed.
- b — Onset/offset timing: baseline delay before absorption.

Findings

- Short-acting insulins are most sensitive to changes in s.
- Long-acting insulins depend more on b (timing constant).
- Helps identify parameters with the greatest impact on glucose control.

Sensitivity guides which parameters need the most precision in modeling.

Why This Matters to Fisheye Software

From Biological Models to Digital Systems

Recreated PK/PD models act as digital twins — virtual representations of human metabolism that allow safe, controlled "what-if" testing before real-world application.

Key Translational Insight

- Translates real-world variables — insulin dose, timing, meals — into computational inputs.
- Same logic applies in software: input variables ⇒ predictive outputs.

Core Technical Skills

- System modeling & numerical simulation
- Uncertainty analysis & model validation
- Data visualization & decision support

Big Picture: *Whether modeling physiological responses or complex engineering systems, the process is that same- iterative simulation, uncertainty quantification, and visualization of outcomes to guide decision-making.*

Takeaways & Q&A

Key Insights

- Rebuilt a **classic PK/PD model** from scratch — almost matched the expected physiological behavior.
- Even with access to the equations and parameters used in a model, a recreation may not be entirely accurate due to outside factors such as inaccessible information or vague parameter definition.
- Demonstrated **reproducibility** and **explainability** through a compact, transparent workflow.
- Framework supports extensions: *meals, feedback control, Sobol analysis, MCMC simulations.*

Thank you!

Questions or thoughts?

References I

[1] H. Zou *et al.*, "Application of pharmacokinetic-pharmacodynamic modeling in drug delivery: Development and challenges," *Frontiers in Pharmacology*, Jun 2020.

[2] H. Derendorf and B. Meibohm, "Modeling of pharmacokinetic/pharmacodynamic (pk/pd) relationships: Concepts and perspectives," *Pharmaceutical Research*, Nov 1999.

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[4] *Glucose Regulation*. University of Guelph / Open Publishing: Pressbooks, –.

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[6] D. M. K. Yadav, "Parameter sensitivity analysis," *Medium*, Dec 2023. Accessed: 2025-11-04.